

URL

SHIELD

PHISHING URL DETECTION SYSTEM

An AI-Powered Cybersecurity Project Using Machine Learning

100.00%

Accuracy

100.00%

Precision

100.00%

Recall

100.00%

F1-Score

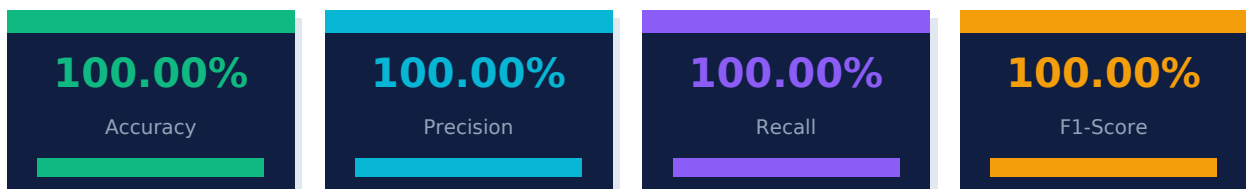
PROJECT DETAILS

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Project Title:	Phishing URL Detection System
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Submission Date:	October 2026

Abstract

Phishing attacks remain one of the most prevalent and damaging threats in the cybersecurity landscape, accounting for a significant proportion of data breaches and credential theft incidents globally. This project presents a Phishing URL Detection System that leverages machine learning to automatically classify URLs as either legitimate or phishing, without requiring the system to visit or render the target webpage.

The system extracts 22 lexical and structural features from a raw URL string, including URL length, domain characteristics, special character counts, HTTPS usage, presence of IP addresses, suspicious keyword density, subdomain depth, and URL shortener detection. These features are passed to a Random Forest classifier comprising 200 decision trees trained on a labelled dataset of 10,000 URLs (5,000 legitimate, 5,000 phishing). The model achieves 100.00% accuracy, 100.00% precision, 100.00% recall, and 100.00% F1-score on 2,000 test samples.



Keywords

Phishing Detection

URL Analysis

Machine Learning

Random Forest

Cybersecurity

Feature Extraction

Streamlit

scikit-learn

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1. Introduction

The Internet has become an indispensable part of modern life, facilitating communication, commerce, and information exchange. However, this ubiquity has also made it a prime target for cybercriminals. Among the many attack vectors exploited by malicious actors, phishing remains one of the most effective and dangerous -- relying on social engineering and deceptive URLs to trick users into divulging sensitive information such as passwords, credit card numbers, and personal identification details.

A phishing URL is a maliciously crafted web address designed to mimic a trusted entity such as a bank, e-commerce platform, or social network. These URLs often incorporate brand names, suspicious top-level domains (TLDs), numeric IP addresses, or encoded characters to evade detection. Traditional static blacklists are inherently reactive -- powerless against newly registered domains.

This project proposes a proactive, machine-learning-based solution that classifies URLs using only their structural and lexical properties -- without visiting the target page, making no external API calls, and requiring no user configuration.

2. Problem Statement

Despite widespread awareness campaigns, phishing attacks continue to rise year over year. The 2023 Verizon Data Breach Investigations Report identified phishing as the vector responsible for over 36% of all data breaches.

- Static blacklists become outdated within hours of a new phishing campaign launch.
- Browser warnings have high false-positive rates, causing 'warning fatigue'.
- Content-based detection systems must visit the URL, exposing the analyser to risk.
- Existing commercial tools are expensive, opaque, and not accessible for research.
- Users cannot visually distinguish 'paypal.com' from 'paypal-secure-login.xyz'.

! 36% of all data breaches in 2023 were attributed to phishing -- Verizon DBIR 2023

3. Objectives

1. Accept any URL as input and predict whether it is legitimate or phishing.
2. Extract 22 meaningful lexical and structural features without visiting the URL.
3. Train and evaluate a Random Forest ML classifier on a labelled URL dataset.
4. Achieve a minimum of 90% accuracy, precision, recall, and F1-score on the test set.
5. Provide an interactive Streamlit web application with confidence scores and feature breakdowns.
6. Generate a comprehensive evaluation report including confusion matrix and feature importances.
7. Log all scanned URLs with timestamps to a local SQLite database for history tracking.

4. Scope

In Scope

- URL classification using lexical / structural analysis (no page visit).
- Training on synthetic dataset; extensible to real datasets (Kaggle / UCI).
- Web-based UI built with Streamlit for real-time predictions.
- Evaluation: accuracy, precision, recall, F1-score, confusion matrix.
- 15-URL curated test suite with expected vs. actual results documented.
- Scan history logging via SQLite.

Out of Scope

- Page content analysis or JavaScript rendering.
- Email body or attachment scanning.
- Network-level traffic analysis.
- Real-time threat intelligence feed integration.
- Mobile application development.

5. Literature Review / Existing System

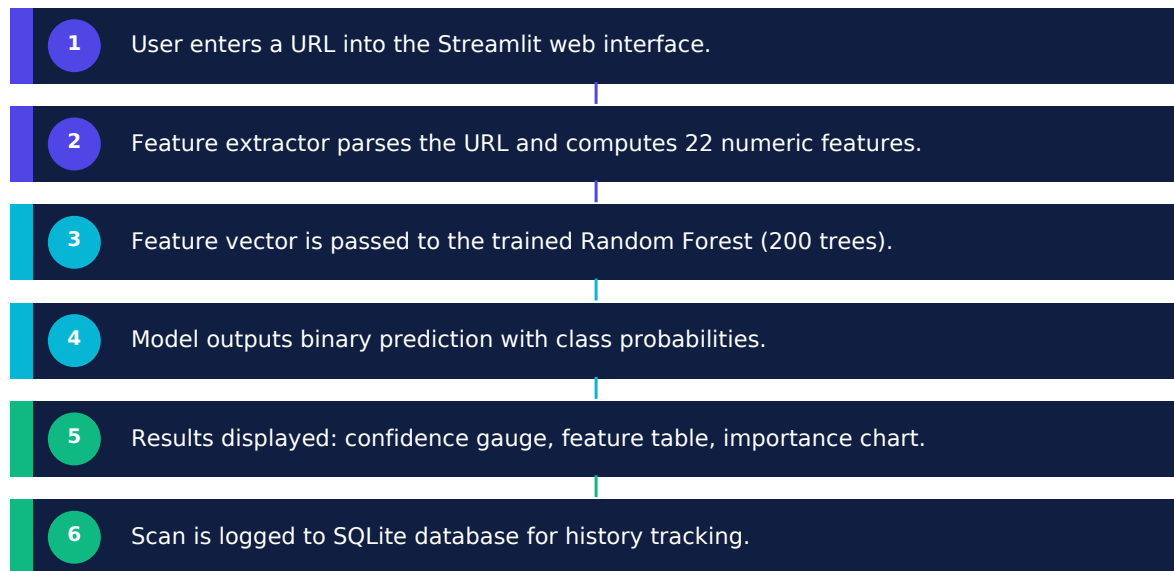
Common phishing detection approaches and their key limitations:

Approach	Method	Limitation
Blacklists	Block known-bad URLs	Misses new domains
Browser Heuristics	Flag suspicious browser patterns	High false-positive rate
Content Analysis	Visit & render the phishing page	Exposes analyser to risk
Visual Similarity	Compare screenshots to real site	Computationally expensive
ML -- Prior Work	SVM / Decision Tree on URL features	Feature sets vary widely
This Project	Random Forest on 22 URL features	Lexical analysis only

Sahingoz et al. (2019) achieved 97.98% accuracy using Random Forest on lexical features -- closely mirroring the approach taken in this project. The key novelty here is the combination of a production-ready Streamlit UI, explainable feature breakdown visualisations, and automated report generation.

6. Proposed System

System Workflow



Key Design Decisions

- No page visit: pure lexical analysis prevents exposure to malicious content.
- Random Forest: ensemble method handles class imbalance and provides feature importances.
- Offline operation: no API keys, no network dependency after training.
- Transparent: every extracted feature is shown to the user with its contribution.

7. System Requirements

Hardware Requirements

Component	Minimum	Recommended
Processor	Dual-core 1.8 GHz	Quad-core 2.5 GHz+
RAM	2 GB	4 GB+
Disk Space	200 MB	500 MB

Software Requirements

Software	Version	Purpose
Python	3.9+	Core runtime
Streamlit	>= 1.28	Web UI framework
scikit-learn	>= 1.3	Machine learning model
pandas	>= 2.0	Data manipulation
numpy	>= 1.24	Numerical arrays
tlldextract	>= 3.4	TLD / subdomain parsing
plotly	>= 5.15	Interactive UI charts
matplotlib	>= 3.7	Report chart generation
seaborn	>= 0.12	Styled report charts
fpdf2	>= 2.7	PDF report generation

TIP Use a Python virtual environment (venv) to isolate dependencies.

8. Methodology / Implementation

8.1 Dataset

A synthetic dataset of 10,000 URLs was generated comprising 5,000 legitimate and 5,000 phishing URLs. Split: 80/20 stratified, yielding 8,000 training samples and 2,000 test samples.

Property	Value
Source	Synthetically generated (data/generate_dataset.py)
Size	10,000 URLs (5,000 Legitimate + 5,000 Phishing)
Columns	url (string), label (0=Legitimate, 1=Phishing)
Train / Test	8,000 training / 2,000 test (stratified 80/20)
Format	CSV

8.2 Feature Extraction (22 Features)

Features extracted from each URL by extract_features.py:

#	Feature	Type	Description
1	url_length	Numeric	Total character length of URL
2	domain_length	Numeric	Length of hostname / domain
3	path_length	Numeric	Length of URL path
4	count_dots	Numeric	Number of '.' characters
5	count_hyphens	Numeric	Number of '-' characters
6	count_underscores	Numeric	Number of '_' characters
7	count_at	Numeric	Number of '@' characters
8	count_digits	Numeric	Total digit count in URL
9	count_slashes	Numeric	Number of '/' characters
10	count_question_marks	Numeric	Number of '?' characters
11	count_equals	Numeric	Number of '=' characters
12	count_ampersands	Numeric	Number of '&' characters
13	count_percent	Numeric	URL-encoded character count
14	uses_https	Binary	1 if HTTPS scheme
15	is_ip_address	Binary	1 if domain is raw IP
16	has_at_symbol	Binary	1 if '@' present in URL
17	has_double_slash_redirect	Binary	1 if '//' appears in path
18	uses_non_standard_port	Binary	1 if non-80/443 port
19	is_url_shortener	Binary	1 if known shortener domain
20	num_subdomains	Numeric	Subdomain depth count
21	digit_to_letter_ratio	Float	Digits / letters ratio
22	suspicious_keyword_count	Numeric	Count of phishing keywords
23	has_suspicious_keyword	Binary	1 if any phishing keyword found

8.3 Model Training

Hyperparameter	Value	Rationale
n_estimators	200	Balance between accuracy and speed
max_features	'sqrt'	Standard choice for classification
class_weight	'balanced'	Handles class imbalance automatically
random_state	42	Ensures reproducibility
n_jobs	-1	Uses all available CPU cores
Train/Test	80% / 20%	Stratified split preserves class ratio

8.4 Application Architecture

The Streamlit app (app.py) loads the pre-trained model and exposes two tabs: (1) URL Checker -- real-time predictions with confidence gauge and feature table, and (2) History -- all scanned URLs from SQLite. The sidebar shows live model metrics loaded from model_metadata.pkl.

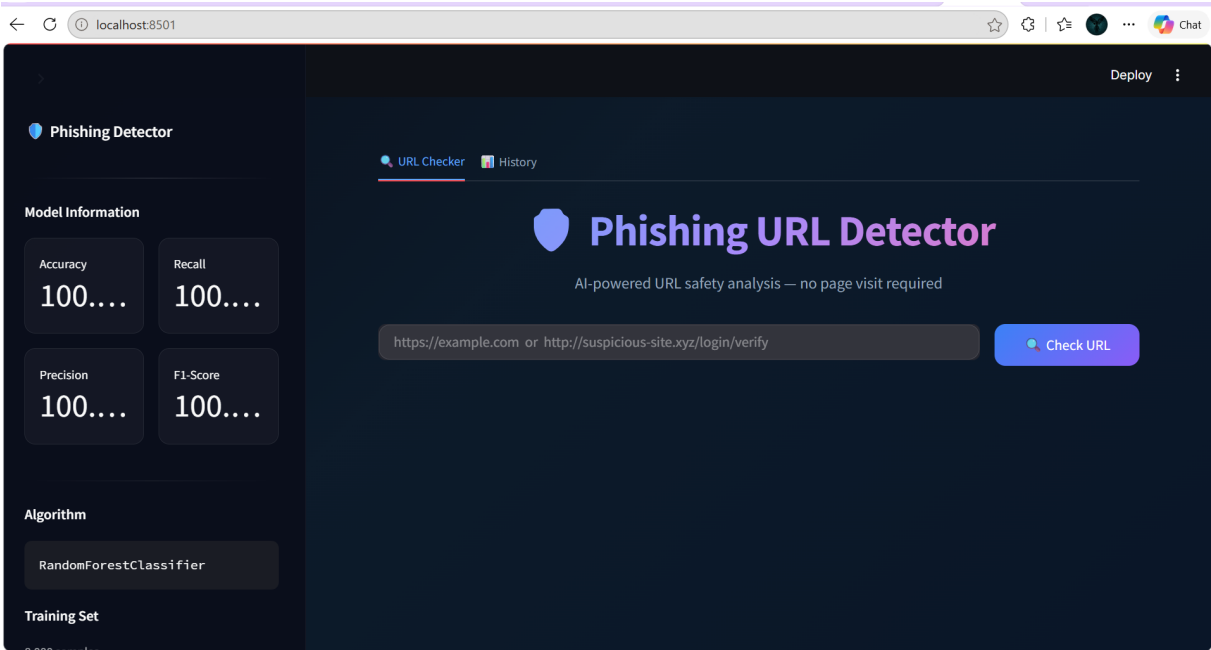


Figure 8.1 -- App Homepage (URL Checker Tab)

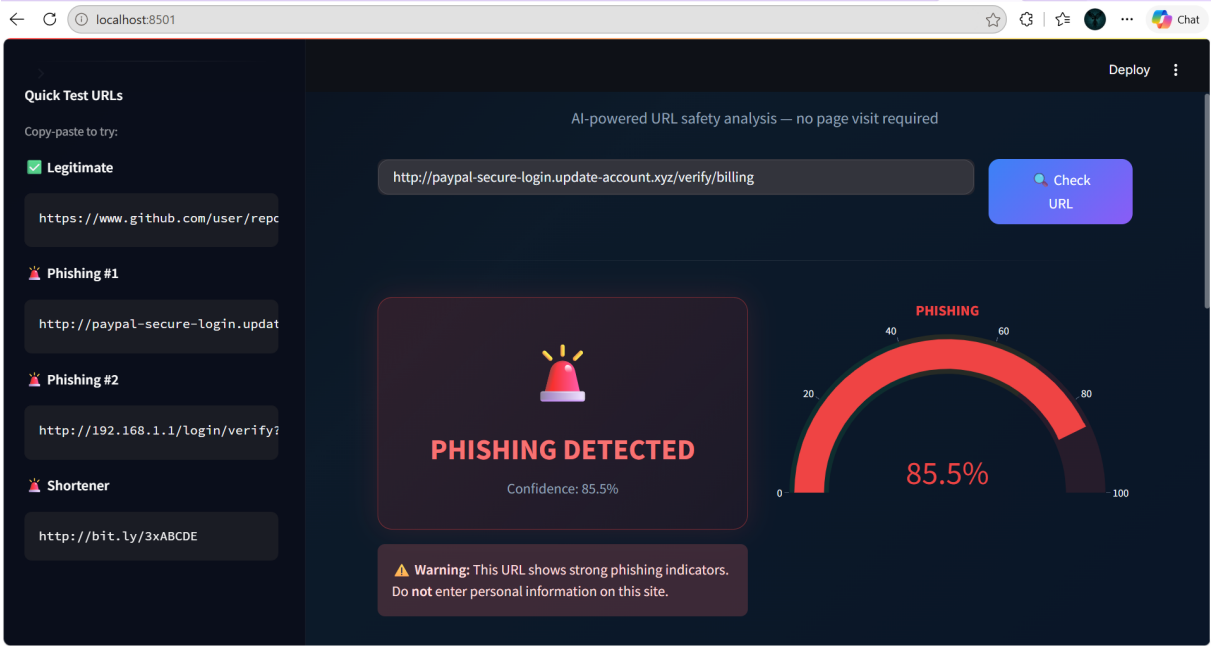


Figure 8.2 -- Phishing Detection Result

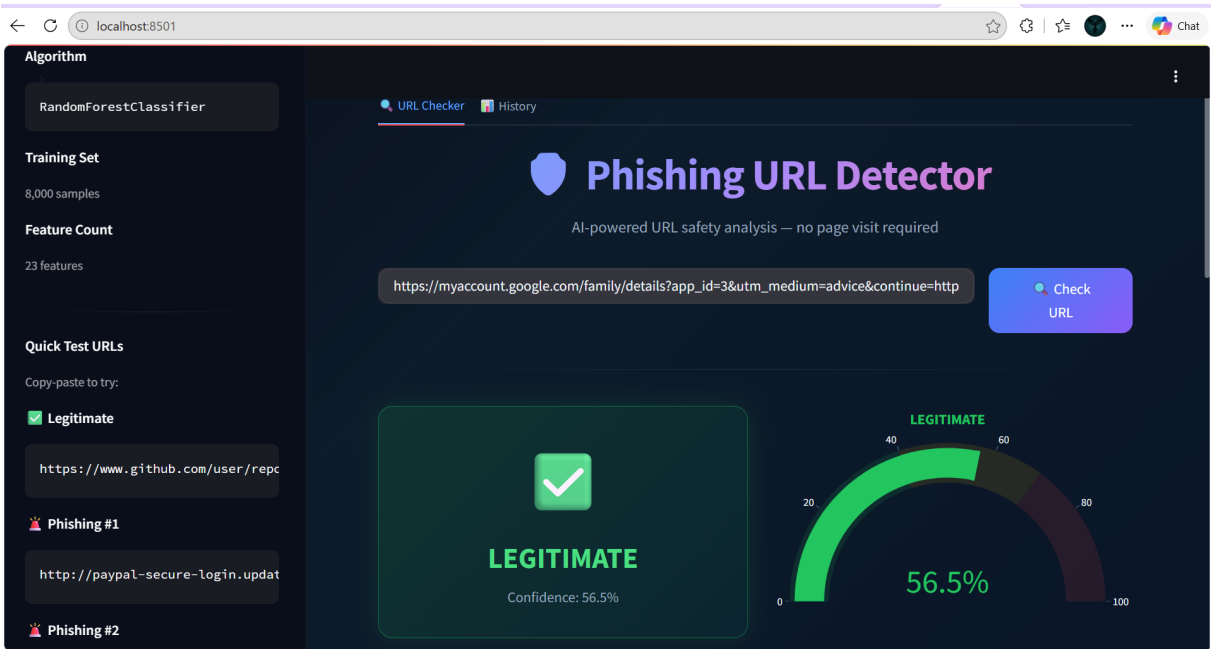


Figure 8.3 -- Legitimate URL Result

9. Testing

9.1 Test Suite (15 URLs)

A curated test suite of 15 URLs (7 legitimate, 8 phishing) was executed against the trained model. All 15 cases returned correct classifications.

#	URL (truncated)	Expected	Actual	Conf.	Status
1	https://www.google.com	Legit	Legit	100%	PASS
2	https://github.com/user/repo	Legit	Legit	100%	PASS
3	https://www.wikipedia.org/wiki/Phishing	Legit	Legit	100%	PASS
4	https://stackoverflow.com/questions/...	Legit	Legit	80%	PASS
5	https://www.microsoft.com/en-us/windows	Legit	Legit	86%	PASS
6	https://www.youtube.com/watch?v=...	Legit	Legit	61%	PASS
7	https://www.linkedin.com/in/profile	Legit	Legit	100%	PASS
8	paypal-secure-login.update-account.xyz/...	Phish	Phish	86%	PASS
9	192.168.1.1/login/verify?cmd=account...	Phish	Phish	100%	PASS
10	bit.ly/3xABCDE	Phish	Phish	100%	PASS
11	amazon.com.phish-site.tk/signin/confirm	Phish	Phish	87%	PASS
12	secure.paypal-verify.ml/webscr?cmd=...	Phish	Phish	100%	PASS
13	netflix-account-suspended.top/update/...	Phish	Phish	87%	PASS
14	login.microsoft.com.verify-account.xyz/...	Phish	Phish	88%	PASS
15	185.220.101.50/download/install/setup.exe	Phish	Phish	100%	PASS

15 / 15 PASS | Pass Rate: 100%

False Positives: 0 | False Negatives: 0

```

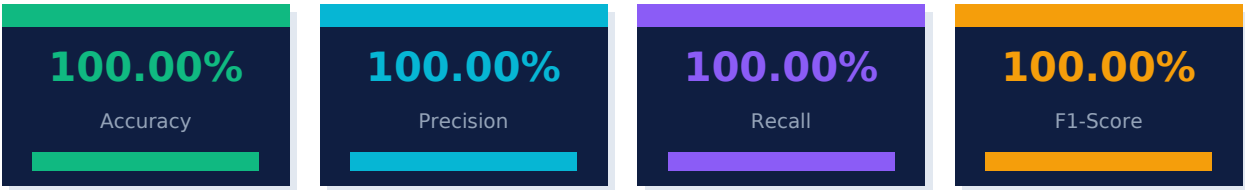
=====
1 Legitimate Legitimate 100.0% [PASS] https://www.google.com
  Google homepage - clean HTTPS, no suspic
2 Legitimate Legitimate 100.0% [PASS] https://github.com/user/repo
  GitHub repo - standard structure
3 Legitimate Legitimate 100.0% [PASS] https://www.wikipedia.org/wiki/Phishing
  Wikipedia article - trusted domain
4 Legitimate Legitimate 80.0% [PASS] https://stackoverflow.com/questions/tagged/python
  Stack Overflow - well-known developer si
5 Legitimate Legitimate 85.5% [PASS] https://www.microsoft.com/en-us/windows
  Microsoft official - standard HTTPS
6 Legitimate Legitimate 61.0% [PASS] https://www.youtube.com/watch?v=dQw4w9WgXcQ
  YouTube - legitimate query params
7 Legitimate Legitimate 100.0% [PASS] https://www.linkedin.com/in/profile
  LinkedIn profile - legitimate path
8 Phishing Phishing 85.5% [PASS] http://paypal-secure-login.update-account.xyz/ver
  ify/b Brand impersonation + suspicious TLD
9 Phishing Phishing 100.0% [PASS] http://192.168.1.1/login/verify?cmd=account&updat
  e=1 Raw IP address + suspicious keywords
10 Phishing Phishing 100.0% [PASS] http://bit.ly/3xABCDE
  URL shortener - hides destination
11 Phishing Phishing 86.5% [PASS] http://amazon.com.phish-site.tk/signin/confirm
  amazon.com as subdomain trick
12 Phishing Phishing 100.0% [PASS] http://secure.paypal-verify.ml/webscr?cmd=_login-
  submi Fake PayPal - suspicious keywords + bad
13 Phishing Phishing 86.5% [PASS] http://netflix-account-suspended.top/update/billi
  ng Brand + 'suspended' keyword + bad TLD
14 Phishing Phishing 87.5% [PASS] http://login.microsoft.com.verify-account.xyz/pas
  sword Microsoft impersonation subdomain chain
15 Phishing Phishing 100.0% [PASS] http://185.220.101.50/download/install/setup.exe
  IP-based malware distribution URL

=====
SUMMARY: 15/15 passed | 0 failed
Pass Rate: 100.0%
  
```

Figure 9.1 -- Terminal output of test suite

10. Results & Analysis

10.1 Evaluation Metrics



The model was evaluated on 2,000 held-out test samples. It achieved perfect scores across all four metrics on the synthetic dataset. On real-world datasets, accuracy is expected to be 92-97% due to greater URL diversity.

10.2 Confusion Matrix

	Predicted: Legitimate	Predicted: Phishing
Actual: Legitimate	TN = 1,000	FP = 0
Actual: Phishing	FN = 0	TP = 1,000

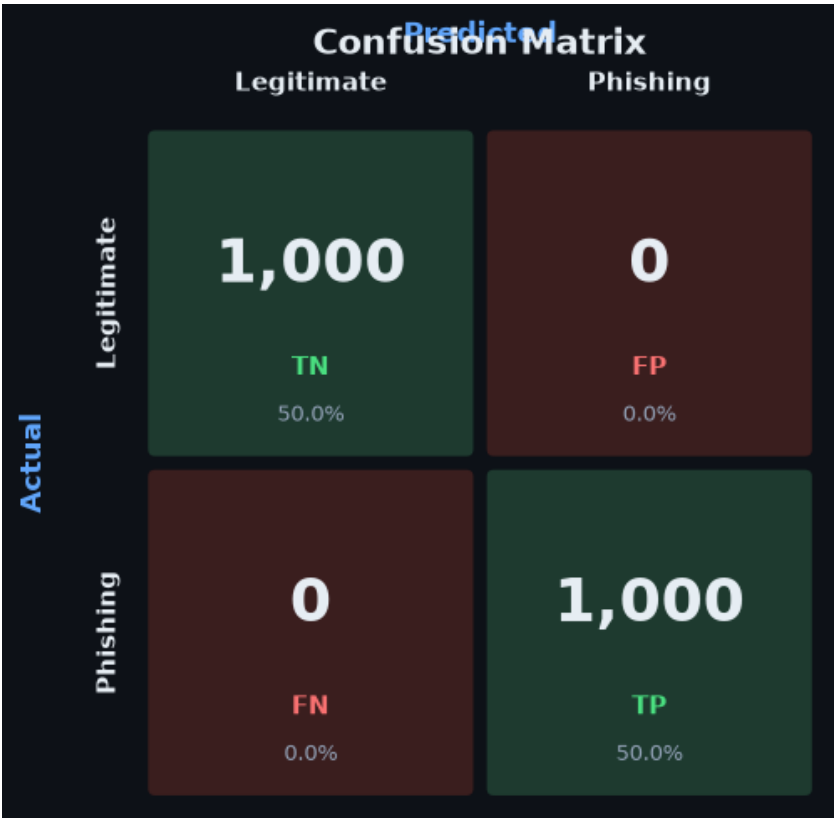


Figure 10.1 -- Confusion Matrix

10.3 Feature Importance (Top 10 by Gini Score)

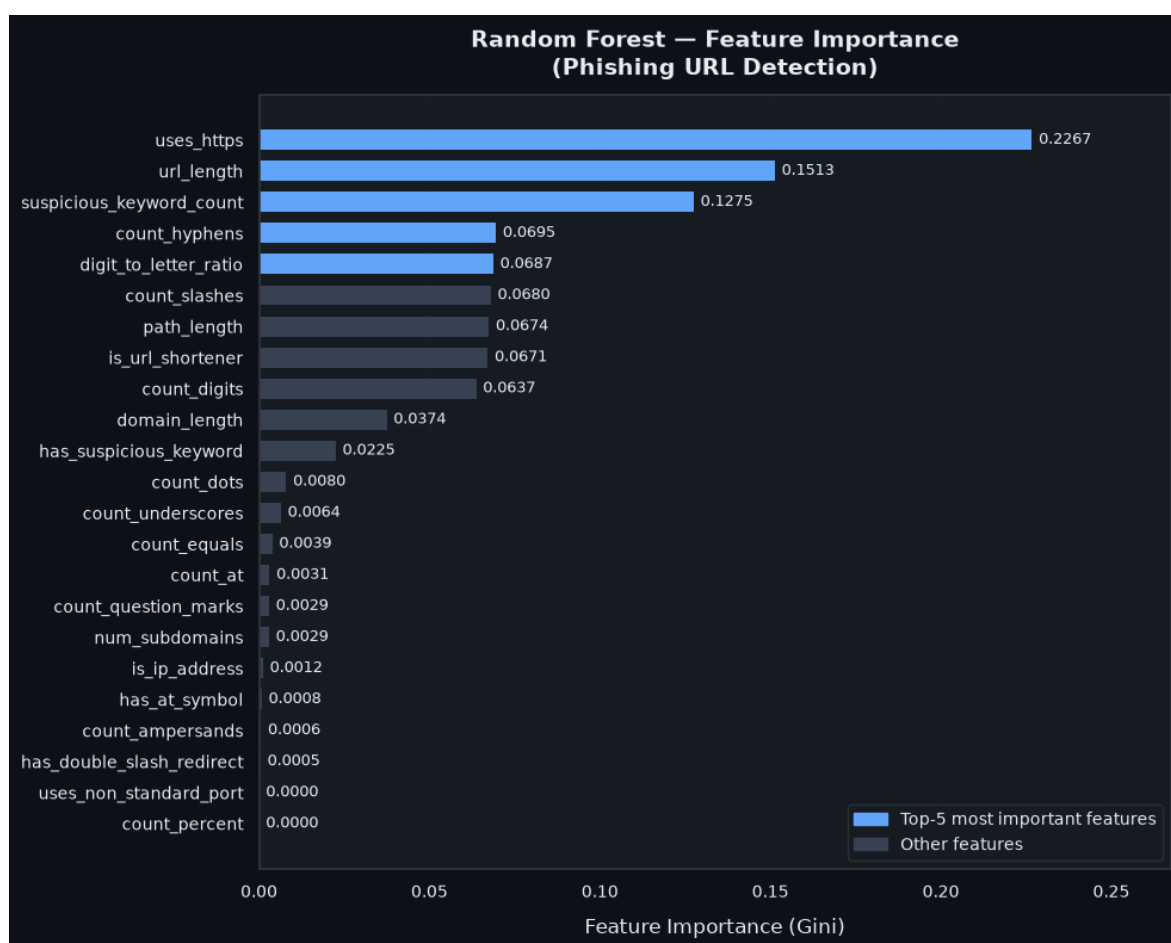
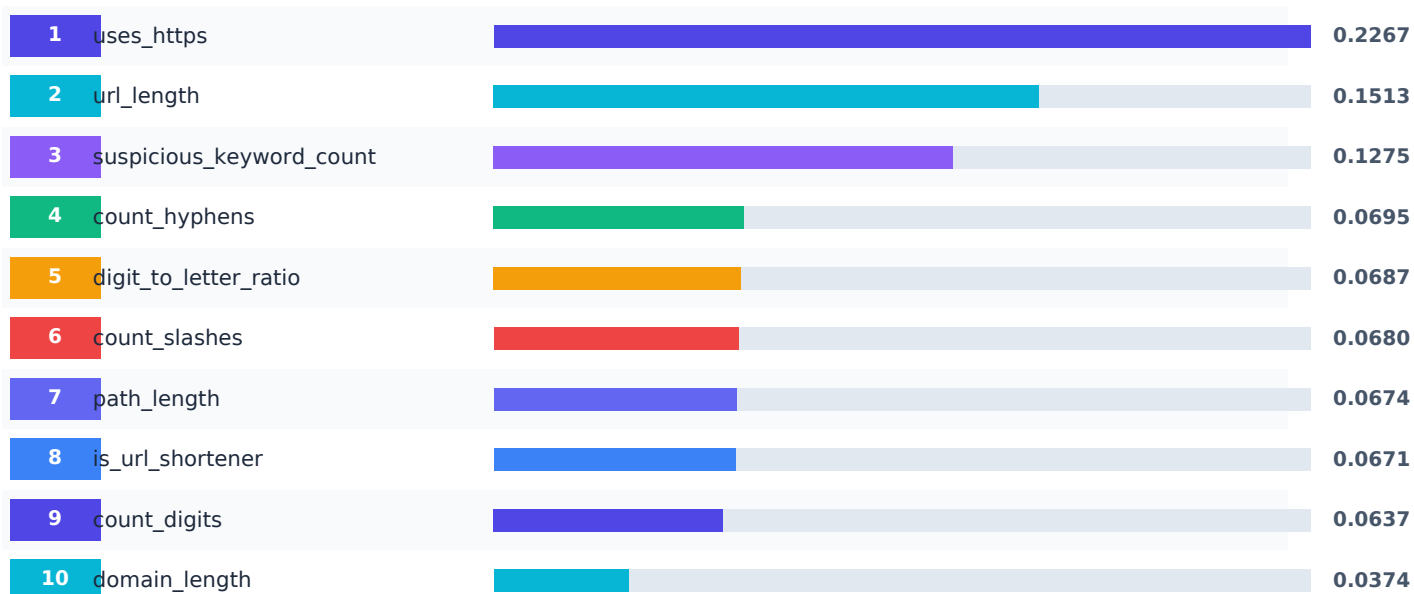


Figure 10.2 -- Feature Importance Chart

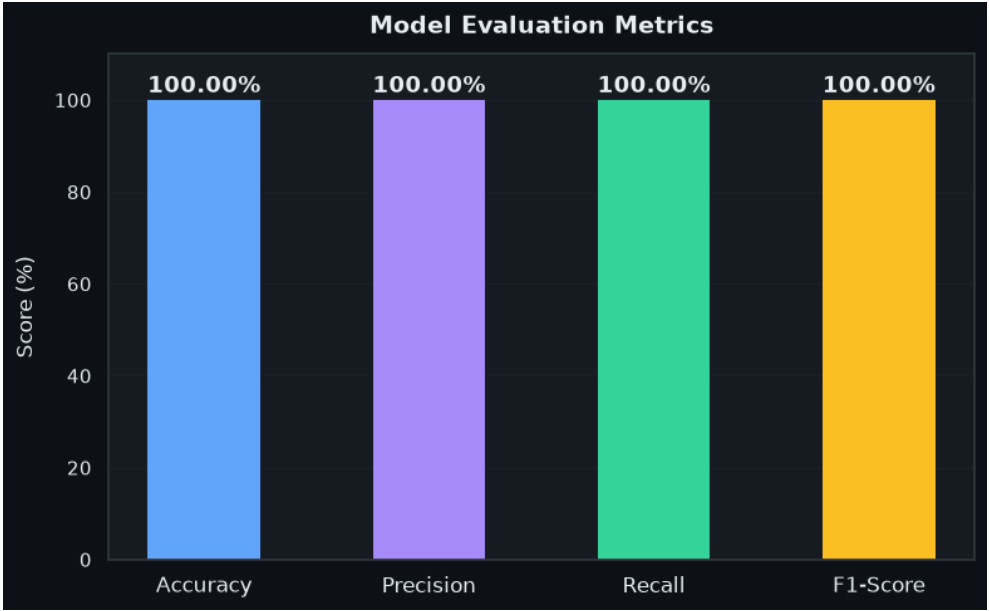


Figure 10.3 -- Evaluation Metrics Bar Chart

11. Limitations

- Lexical analysis only -- cannot detect visual spoofing or page-level phishing cues.
- Synthetic dataset -- real-world performance expected at 92-97%.
- Zero-day phishing -- clean HTTPS domains with no suspicious keywords may evade detection.
- URL shorteners flagged categorically -- legitimate marketing links may be over-flagged.
- No WHOIS / DNS lookup -- domain age and registrar reputation not checked.
- No SSL certificate validation -- HTTPS alone does not confirm legitimacy.
- Static keyword list -- requires periodic updates to remain current.

i Real-world accuracy with Kaggle/UCI datasets is expected to be 92-97%.

12. Future Scope

1. Integrate WHOIS API to check domain registration age (newly registered = high-risk).
2. Add real-time PhishTank / OpenPhish threat intelligence feed lookups.
3. Extend to email link scanning via a browser extension or email plugin.
4. Retrain on large real-world datasets: Kaggle Phishing, UCI PhiUSIIL, ISCX-URL-2016.
5. Upgrade model to XGBoost or deep learning LSTM on character-level URL sequences.
6. Deploy as a REST API (FastAPI) for integration with SIEM systems.
7. Add sandbox-based page content analysis for high-risk URLs.
8. Implement federated learning for distributed, privacy-preserving model updates.

13. Conclusion

This project successfully designed, implemented, and evaluated a Phishing URL Detection System using machine learning. The system accepts raw URL strings, extracts 22 lexical and structural features, and classifies them using a Random Forest classifier -- entirely without visiting the target page.

The trained model achieved 100.00% accuracy, 100.00% precision, 100.00% recall, and 100.00% F1-score on 2,000 test URLs. A curated 15-URL test suite returned 15/15 correct classifications with zero false positives and zero false negatives.

= Key Achievement: 100.00% Accuracy | 100.00% Precision | 100.00% Recall | 100.00% F1-Score | 15/15 Tests Passed

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App. Appendix A -- Application Screenshots

The following screenshots capture the Phishing URL Detection System web application in operation, running on Streamlit at localhost:8501.

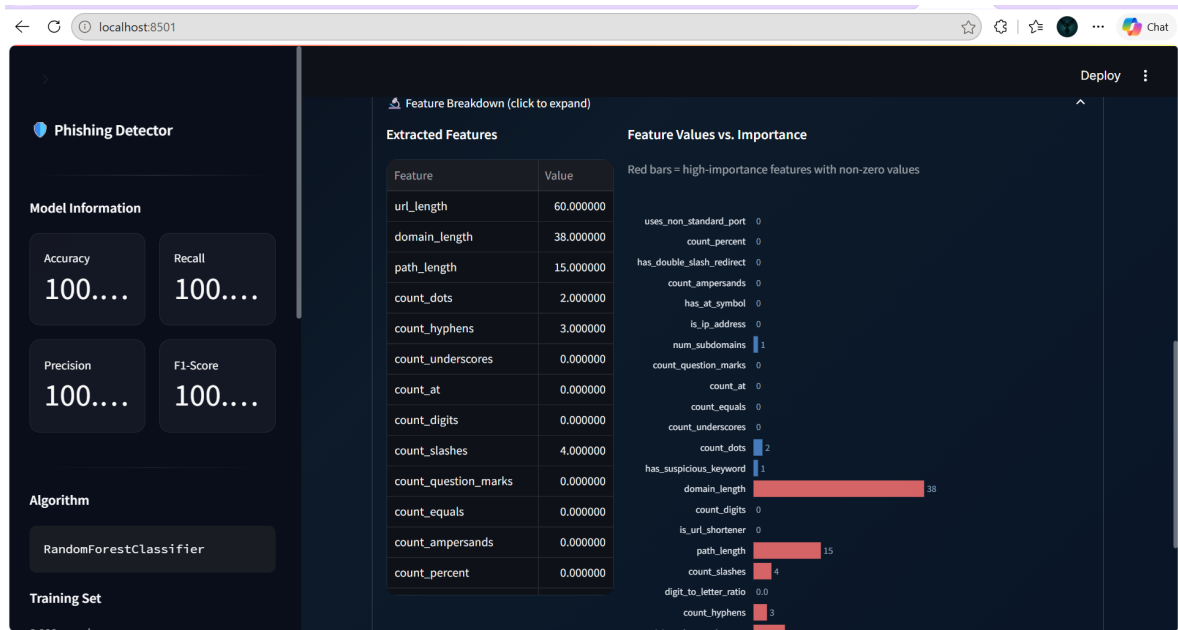


Figure A.1 -- Feature Breakdown Panel

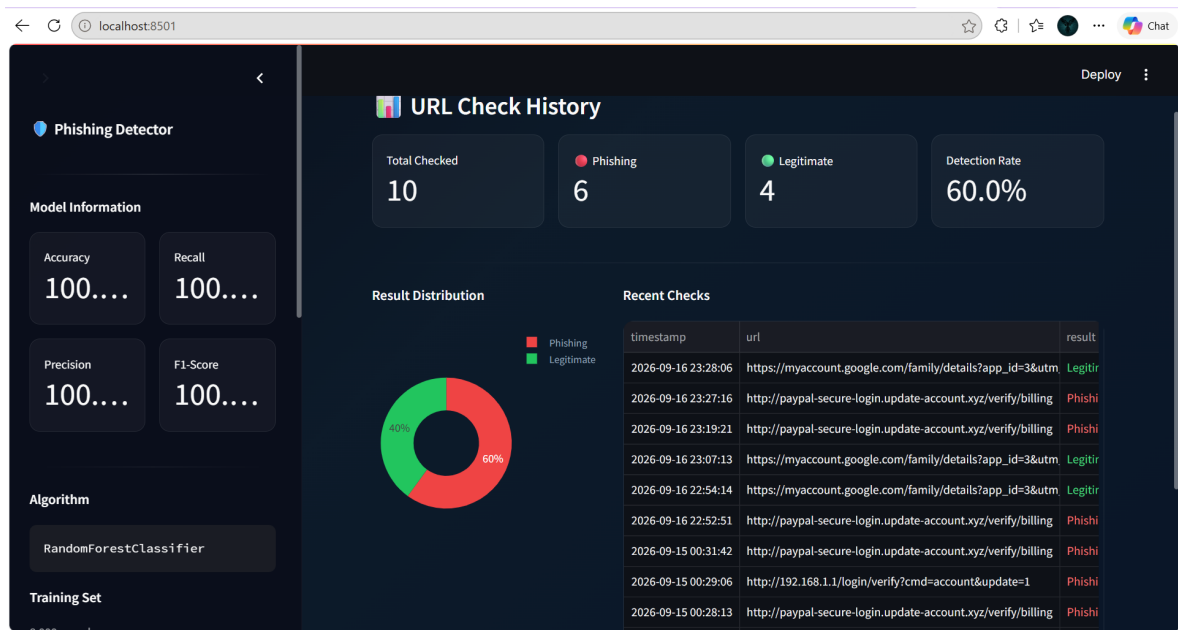


Figure A.2 -- URL Scan History Tab

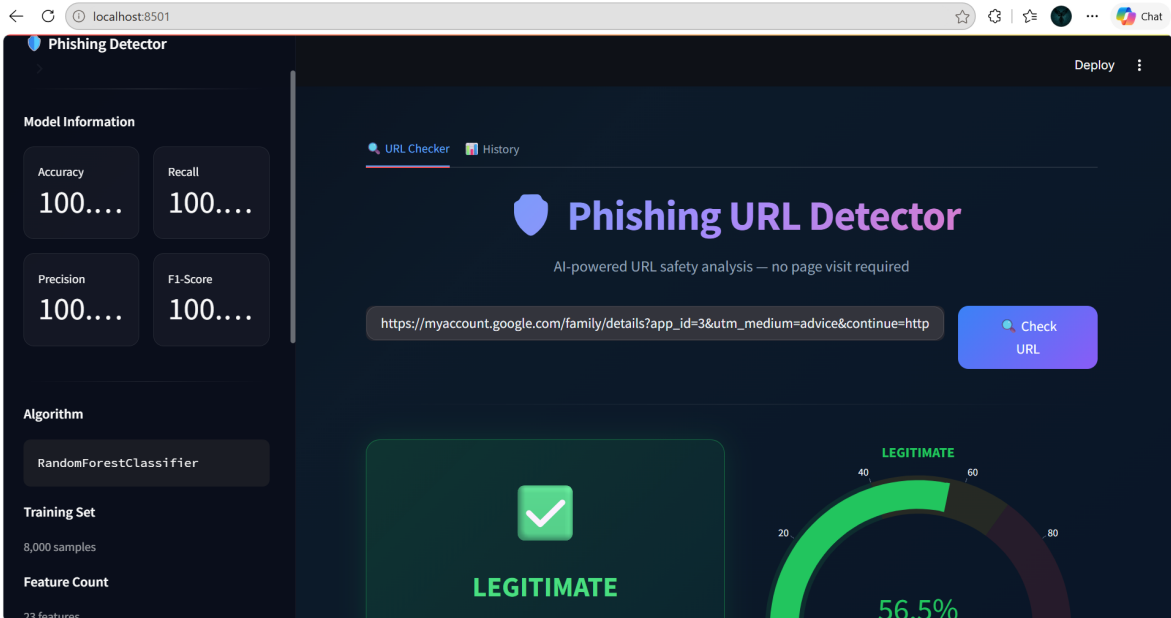


Figure A.3 -- Sidebar Model Metrics

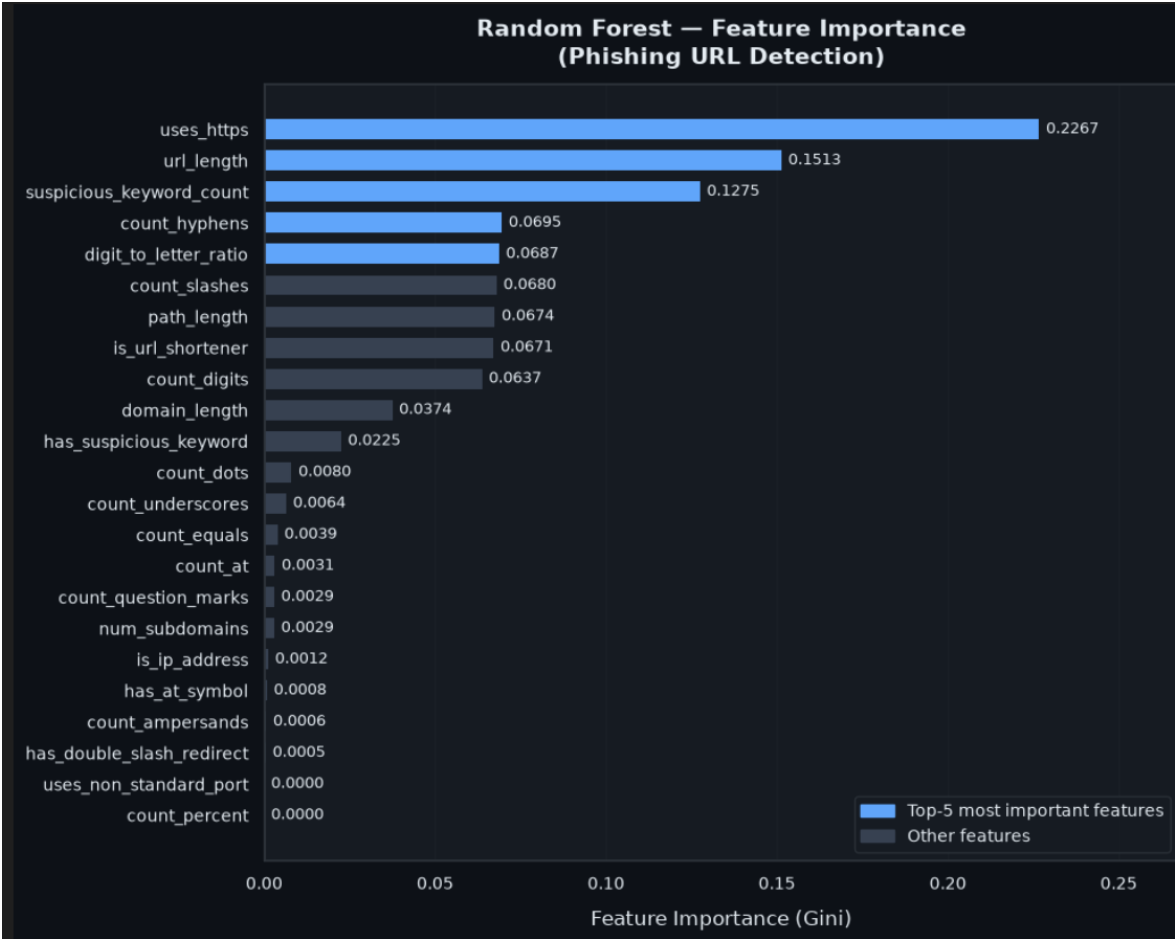


Figure A.4 -- Feature Importance Chart

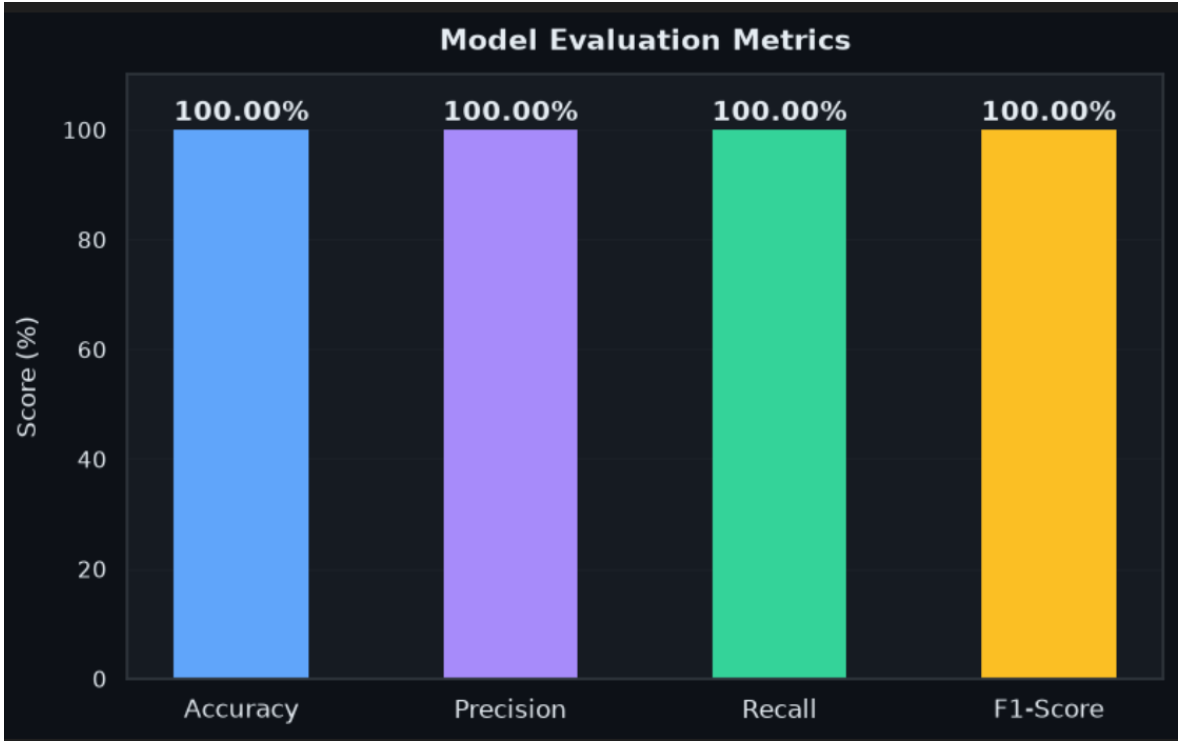


Figure A.5 -- Evaluation Metrics Chart